Guru Nanak Dev University, Amritsar

Department of CET

Practical file of

Engineering Graphics and Drafting

Using

AutoCad

Submitted to:

Er. Amrinder Singh

Assistant Professor

Department of Mechanical Engineering

Submitted by:

Karman Singh Sethi

17032206379

Practical

1. Lab: Auto CAD Lab
2. Title of the Practical: Study and draw 2-D sketching entities like lines, rectangle, parallelogram polygon, circle etc., under SKETCH ENTITY MENU
3. Teacher Name: Er. Amrinder Singh
4. AIM: Study and draw 2-D sketching entities like lines, rectangle, parallelogram polygon, circle etc., under SKETCH ENTITY MENU
5. Apparatus/Software used with specifications: **AUTO CAD Software**
6. Major Components used: **AUTO CAD Software**
7. Theory:

A drawing is prepared traditionally using instruments. The modern alternative is to draw with the aid of computers. This method is known as computer aided drafting (or) Computer aided drawing (or) computer assisted drafting (or) computer argument drafting and computer automated drafting. Modern CAD systems are based on the interactive computer graphics. Interactive computer graphics denotes a user defined system in which the computer is employed to create and display data in the terms of pictures and symbols, by the communication between the user and the system. A typical interactive computer graphics system in combination of both hardware and software. The four fundamental elements of drawing are points, straight lines, arcs and circles. In computer aided drawing systems, the points on the screen be located by selected with the cursor, snapping to the grip, points (or) by entering the numerical values of the co-ordinates of the points. The other elements can be drawn easily by using relevant commands.

Command	**Description**	**Options**
LINE or L	Draws straight lines	<RET> **In reply to From Point:** prompt, line begins at end of previous line or arc **C In reply to To point:** prompt, closes the polygon back to first "From Point" **U In reply to To point:** prompt, undoes last line segment

POLYGON	Draws regular polygons with a specified number of sides. Polygons are Polyline entities.	E Specifies size and rotation of polygon by picking endpoints of one edge C Circumscribes polygon around a circle I Inscribes polygon within a circle
RECTANGLE	Draws Rectangle with specified dimension of sides. Rectangles are polyline entities	
CIRCLE or C	Draws a circle of any size. The default method is to pick a center point and pick a point on the radius or type the radius dimension, but other methods can be selected.	2P Specifies circle by picking 2 points on the diameter 3P Specifies circle by picking 3 points through which the circle will pass D Allows entering the diameter dimension instead of radius dimension TTR Specifies circle by picking two lines, arcs or circles for the circle to be tangent to, and entering the dimension of the radius <RET> Enters radius of circle (the default)

8. **Procedure /Algorithm:**

Command: _ Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter] <25.0000>: D

Specify diameter of circle <50.0000>: 20

Command: Polygon

Command: _polygon Enter number of sides <4>: 5

Specify center of polygon or [Edge]:

Enter an option [Inscribed in circle/Circumscribed about circle] <I>: C

Specify radius of circle: 10:

Command: _Rectangle

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]:

Specify other corner point or [Area/Dimensions/Rotation]: D

Specify length for rectangles <10.0000>: 20

Specify width for rectangles <10.0000>: 16

Specify other corner point or [Area/Dimensions/Rotation]:

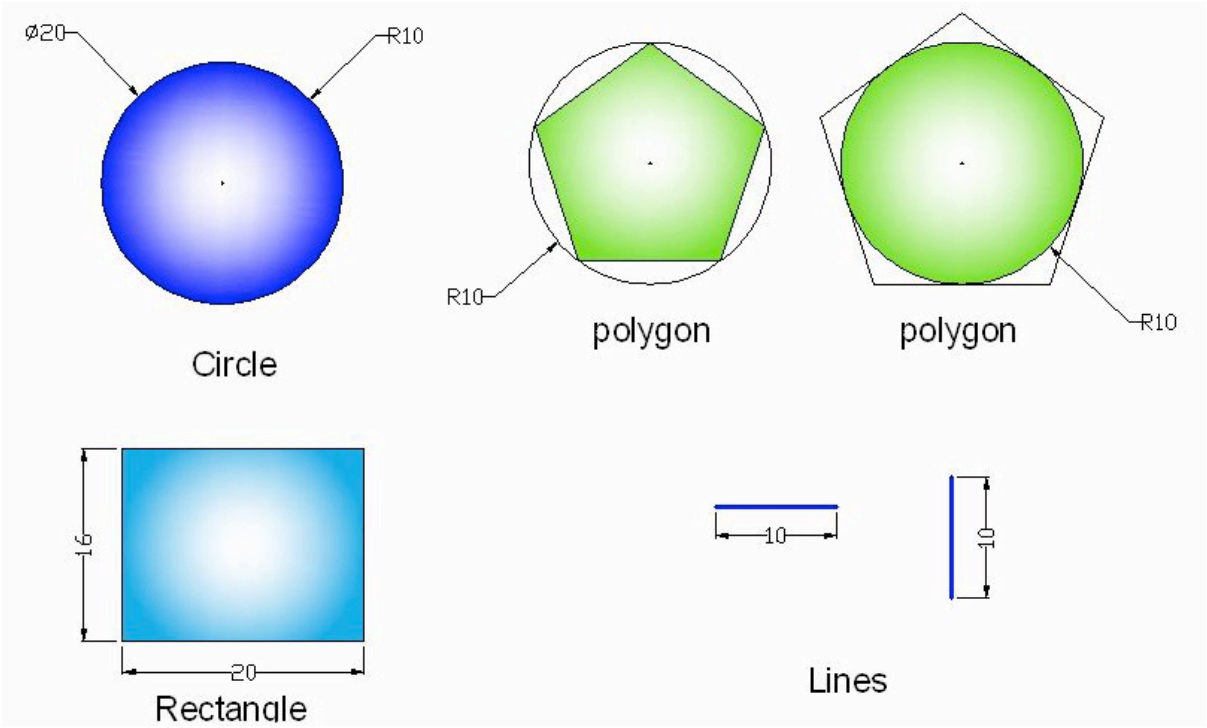
Command:_Line

Command: _line Specify first point:

Specify next point or [Undo]: <Ortho on> 10

Enter

9. Diagrams:



10. Output & Results:

Output: Students will understand the various methods to draw circle, rectangle, line, and polygon.

Result:- The circle, polygon, rectangle and lines are drawn according to dimensions.

Practical

1. Lab: Auto CAD Lab

2. Title of the Practical: Rectangular array & Polar array

3. Teacher Name: Er. Amrinder Singh

4. AIM: Rectangular array & Polar array

5. Apparatus/Software used with specifications: **AUTO CAD Software**

6. Major Components used: **AUTO CAD Software**

7. Theory:

ARRAY creates multiple copies of objects in a pattern. ARRAY is of two types: i.e. Rectangular ARRAY and Polar ARRAY.

Rectangular ARRAY:- In this command we can create multiple copies in the form of rows and columns. We can specify number of rows and columns according to requirement. We can also specify row distance and column distance.

Polar ARRAY:- In this command we can create multiple copies along the centre of a circle. We can specify number of objects, angle between objects and angle to fill.

8. Procedure /Algorithm:

Command: Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter]: D

Specify diameter of circle: 20

Command: Ploygon

Command: _polygon Enter number of sides <4>: 5

Specify center of polygon or [Edge]:

Enter an option [Inscribed in circle/Circumscribed about circle] <I>: C

Specify radius of circle: 10

Command: _Array

Select objects: 1 found

Select objects:

ok

Command: Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter] <225.0000>: D

Specify diameter of circle <450.0000>: 50

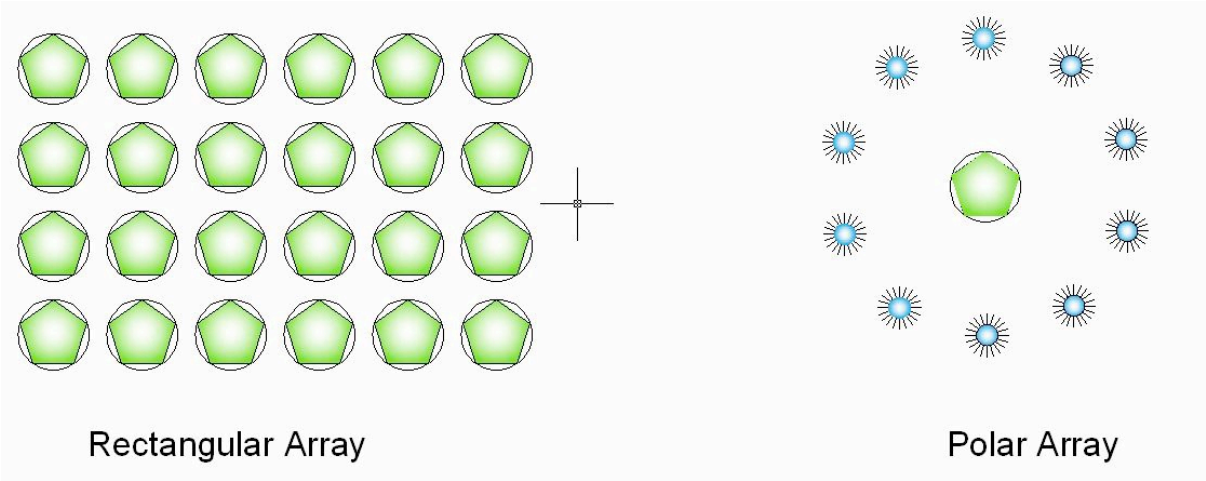
Command: _Array

Select objects: 1 found

Select objects:

Ok

9. Diagrams:



10. Output & Results:

Output: Students will understand the various methods to draw the multiple copies of a single entity with the help of array command.

Result:- The rectangular array and polar array was used to draw the multiple copies of the given entity.

11. Industrial Applications of the said practical: :

These commands will be used to draw 2D Drawings in various industries.

Practical

1. Lab: CG Lab
2. Title of the Practical: Sketch of Metal grate
3. Teacher Name: Er. Amrinder Singh
4. AIM: To draw the sketch of Metal grate
5. Apparatus/Software used with specifications: **AUTO CAD Software**
6. Major Components used: **AUTO CAD Software**
7. Theory:

A grating is any regularly spaced collection of essentially identical, parallel, elongated elements. Gratings usually consist of a single set of elongated elements, but can consist of two sets, in which case the second set is usually perpendicular to the first (as illustrated).[1] When the two sets are perpendicular, this is also known as a grid or a mesh.

8. **Procedure /Algorithm:**

Command: _Rectangle

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]:

Specify other corner point or [Area/Dimensions/Rotation]: D

Specify length for rectangles <10.0000>: 60

Specify width for rectangles <10.0000>: 25

Command: _offset

Specify offset distance or [Through/Erase/Layer] <Through>: 5

Select object to offset or [Exit/Undo] <Exit>:

Specify point on side to offset or [Exit/Multiple/Undo] <Exit>:

Command: _Rectangle

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]:

Specify other corner point or [Area/Dimensions/Rotation]: D

Specify length for rectangles <10.0000>: 10

Specify width for rectangles <10.0000>: 5

Command: Extrude

Select object to extrude

Specify height of extrusion or [Direction/Path/Taper angle]: 50

Command: SUBTRACT

Select solids, surfaces, and regions to subtract from

Select objects: 1 found

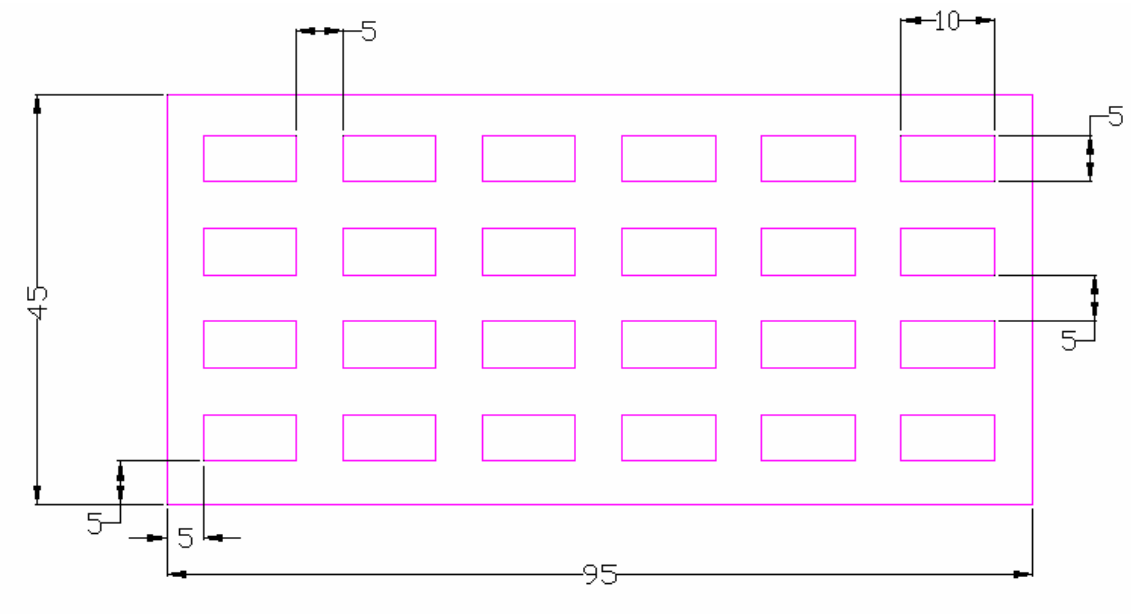
Select objects:

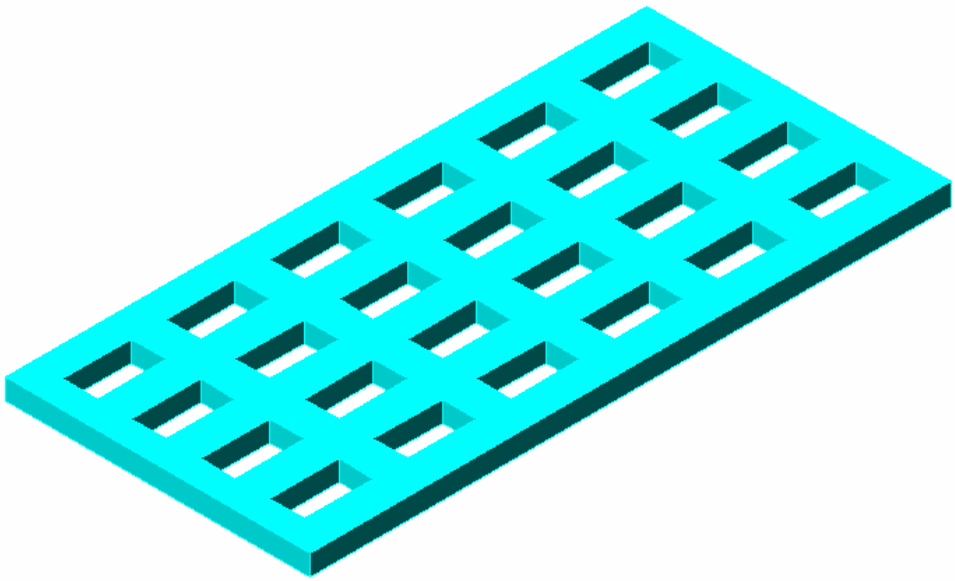
Select solids, surfaces, and regions to subtract ..

Select objects: 24found

Enter

9. Diagrams:





10. Output & Results:

Output: Students will be able to understand the use of various draw and modify commands

Result:- The metal grate was drawn according to the given dimensions.

11. Industrial Applications of the said practical: :

These commands will be used for drafting of metal grate in various industries.

Practical

1. Lab: CG Lab
2. Title of the Practical: Sketch of Slotted Base
3. Teacher Name: Er. Amrinder Singh
4. AIM: To draw the sketch of Slotted Base
5. Apparatus/Software used with specifications: **AUTO CAD Software**
6. Major Components used: **AUTO CAD Software**
7. Theory:

An slotted base is a work holding device used as a fixture in metalworking.

The slotted base is made from high quality material (generally spheroidal cast iron) that has been stabilized to prevent further movement or distortion. Slotted holes or *T* bolt slots are machined into the surfaces to enable the secure attachment or clamping of workpieces to the plate, and also of the plate to the worktable.

Slotted base also may be used to hold the workpiece square to the table during marking out operations.

8. Procedure /Algorithm:

Command: _Rectangle

Specify first corner point or [Chamfer/Elevation/Fillet/Thickness/Width]:

Specify other corner point or [Area/Dimensions/Rotation]: D

Specify length for rectangles <10.0000>: 60

Specify width for rectangles <10.0000>: 25

Command: _ Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter] <25.0000>: D

Specify radius of circle <50.0000>: 5

Command: _offset

Specify offset distance or [Through/Erase/Layer] <Through>: 3

Select object to offset or [Exit/Undo] <Exit>:

Specify point on side to offset or [Exit/Multiple/Undo] <Exit>:

Command: Extrude

Select object to extrude

Specify height of extrusion or [Direction/Path/Taper angle]: 50

Command: SUBTRACT

Select solids, surfaces, and regions to subtract from

Select objects: 1 found

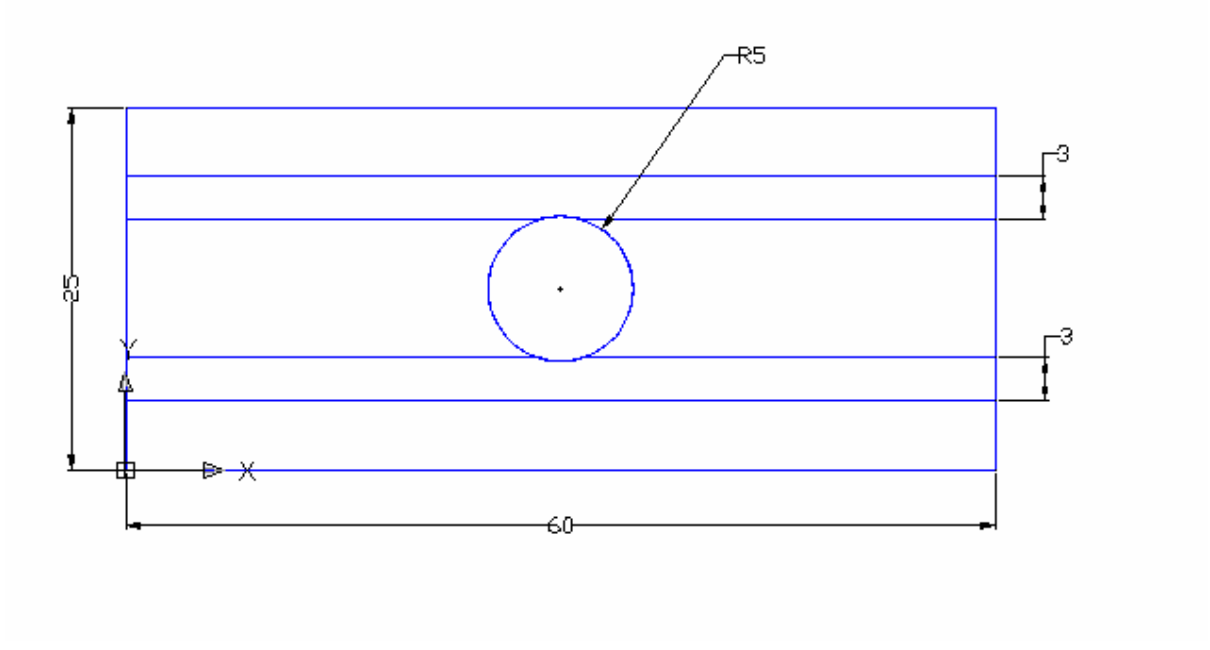
Select objects:

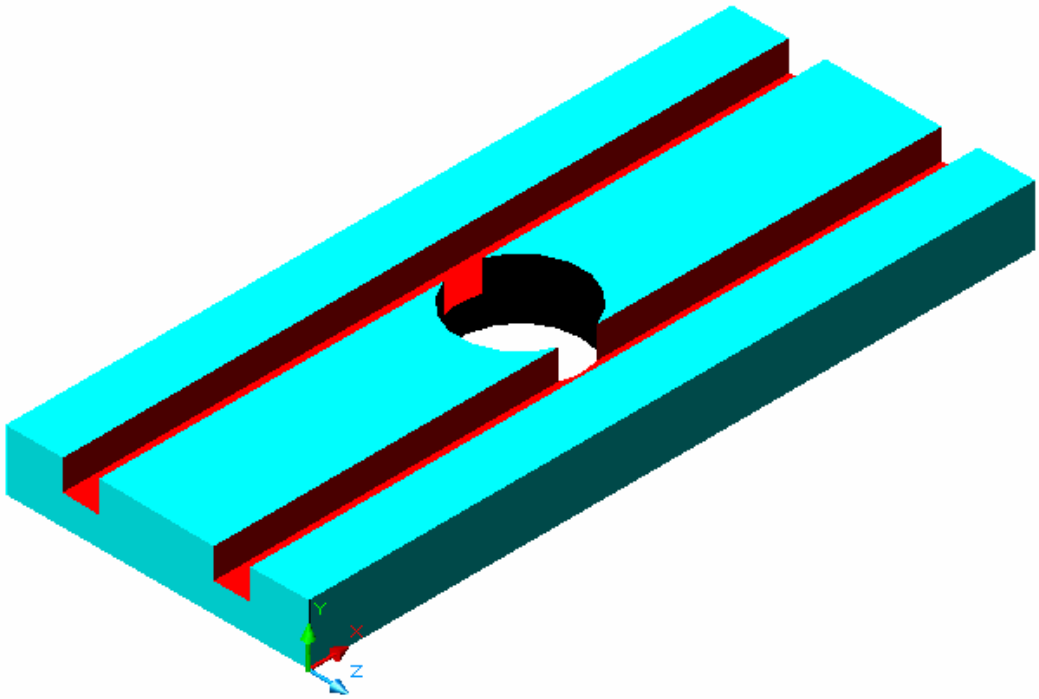
Select solids, surfaces, and regions to subtract ..

Select objects: 3 found

Enter

9. Diagrams:





10. Output & Results:

Output:

1. Students will be able to understand the use of various draw and modify commands.
2. Students will be able to understand the use of Extrude command.
3. Students will be able to understand the use of Subtract command.

Result:- The slotted base was drawn according to the given dimensions.

11. Industrial Applications of the said practical: :

These commands will be used for drafting of slotted base in various industries.

Practical

1. Lab: CG Lab
2. Title of the Practical: Sketch of Link
3. Teacher Name: Er. Amrinder Singh
4. AIM: To draw the sketch of Link
5. Apparatus/Software used with specifications: **AUTO CAD Software**
6. Major Components used: **AUTO CAD Software**
7. Theory:

A **mechanical linkage** is an assembly of bodies connected to manage forces and movement. The movement of a body, or link, is studied using geometry so the link is considered to be rigid.[1] The connections between links are modeled as providing ideal movement, pure rotation or sliding for example, and are called joints.

Linkages may be constructed from open chains, closed chains, or a combination of open and closed chains. Each link in a chain is connected by a joint to one or more other links.

8. **Procedure /Algorithm:**

Command: _ Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter] <25.0000>: r

Specify radius of circle <50.0000>: 10

Command: _ Circle

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]:

Specify radius of circle or [Diameter] <25.0000>: r

Specify radius of circle <50.0000>: 5

Command: _offset

Specify offset distance or [Through/Erase/Layer] <Through>: 30

Select object to offset or [Exit/Undo] <Exit>:

Specify point on side to offset or [Exit/Multiple/Undo] <Exit>:

Command: Tan Tan radius

Command: _circle Specify center point for circle or [3P/2P/Ttr (tan tan radius)]: _ttr

Specify point on object for first tangent of circle:

Specify point on object for second tangent of circle:

Specify radius of circle <10.0000>: 15

Command: Extrude

Select object to extrude

Specify height of extrusion or [Direction/Path/Taper angle]: 50

Command: SUBTRACT

Select solids, surfaces, and regions to subtract from

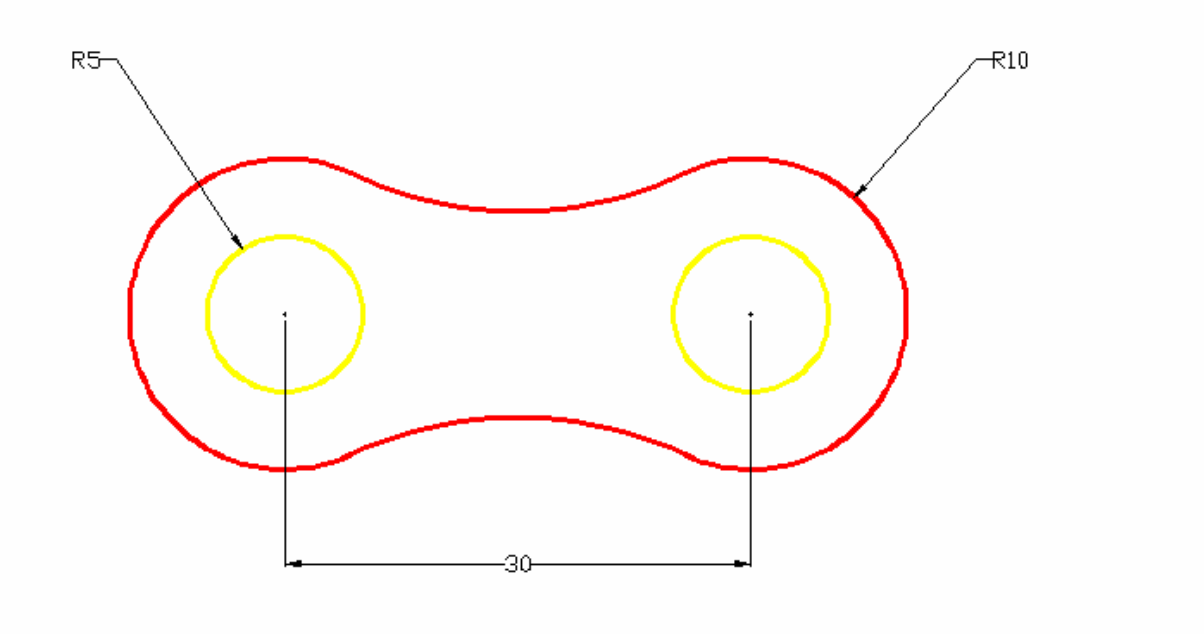
Select objects: 1 found

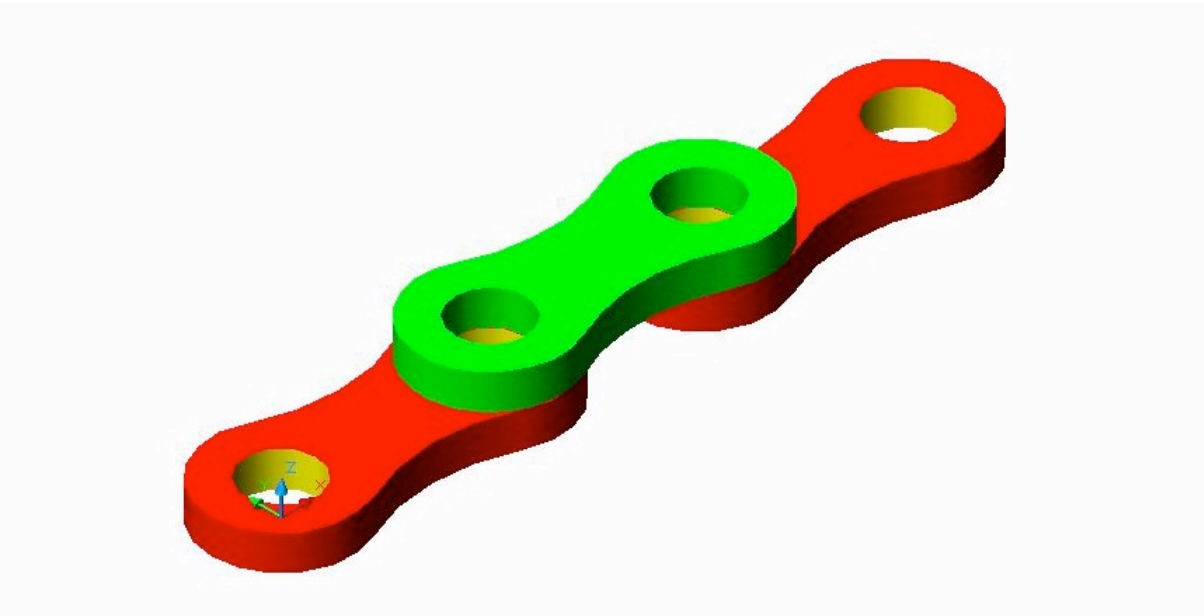
Select objects:

Select solids, surfaces, and regions to subtract ..

Select objects: 3 found, Enter

9. Diagrams:





10. Output & Results:

Output: 1. Students will be able to understand the use of various draw and modify commands

2. Students will be able to understand the use of Extrude command.

Result:- The slotted base was drawn according to the given dimensions.

11. Industrial Applications of the said practical: :

These commands will be used for drafting of slotted base in various industries.